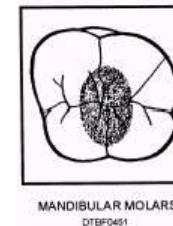
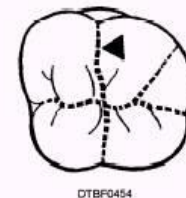
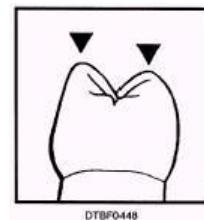
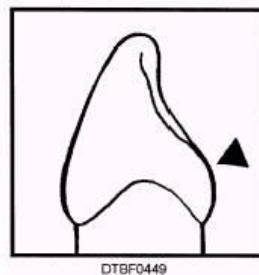
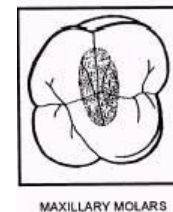
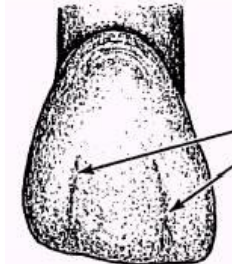
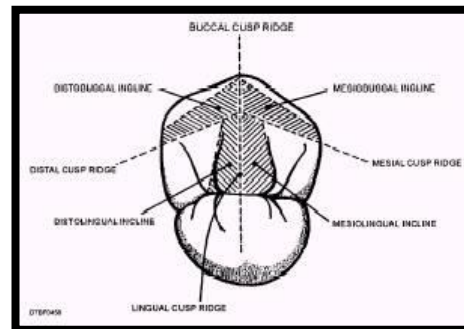
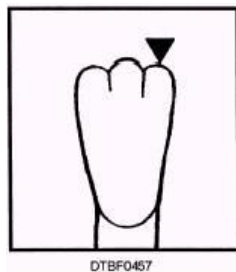


Surface anatomy

- **Surface anatomy:** The tooth surface is not a flat surface. There are elevations and depressions. Each elevation and depression is described based on its location, shape and function.



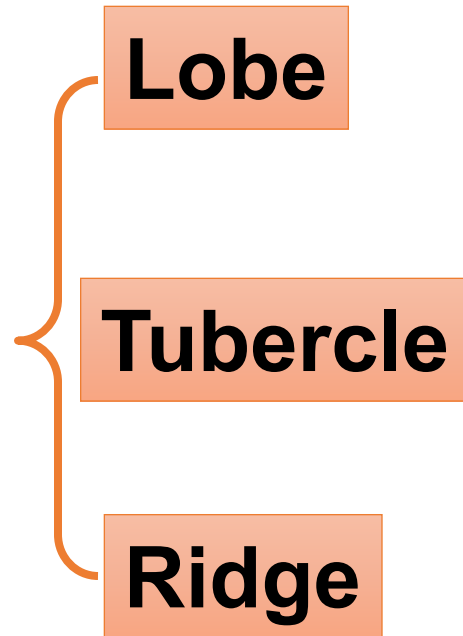
Anatomical landmarks of Tooth crown

Crown Elevations

Lobe

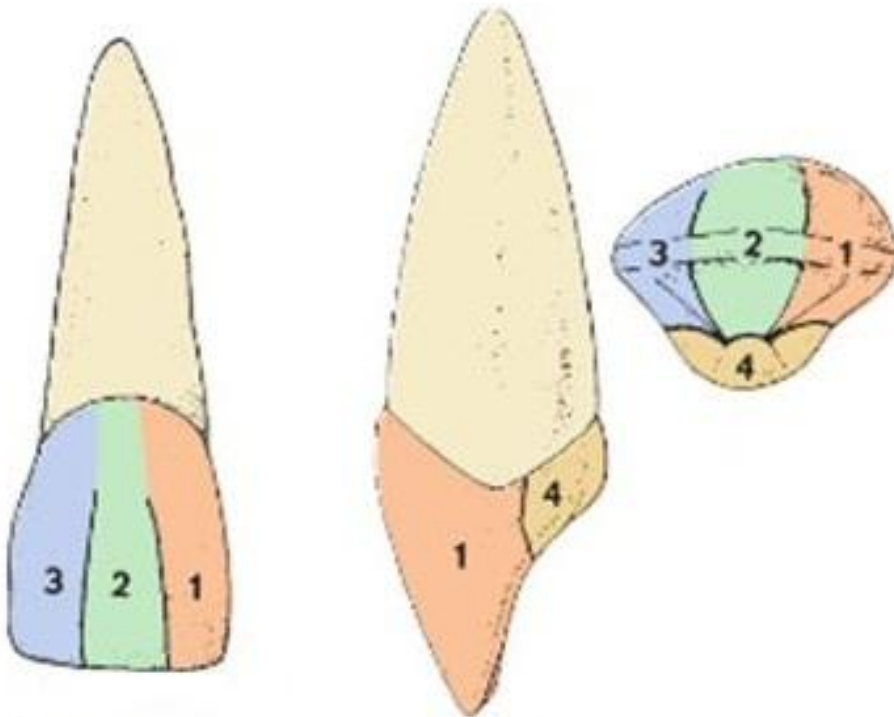
Tubercle

Ridge



Lobe

One of the primary centers of calcification and growth formed during crown development.

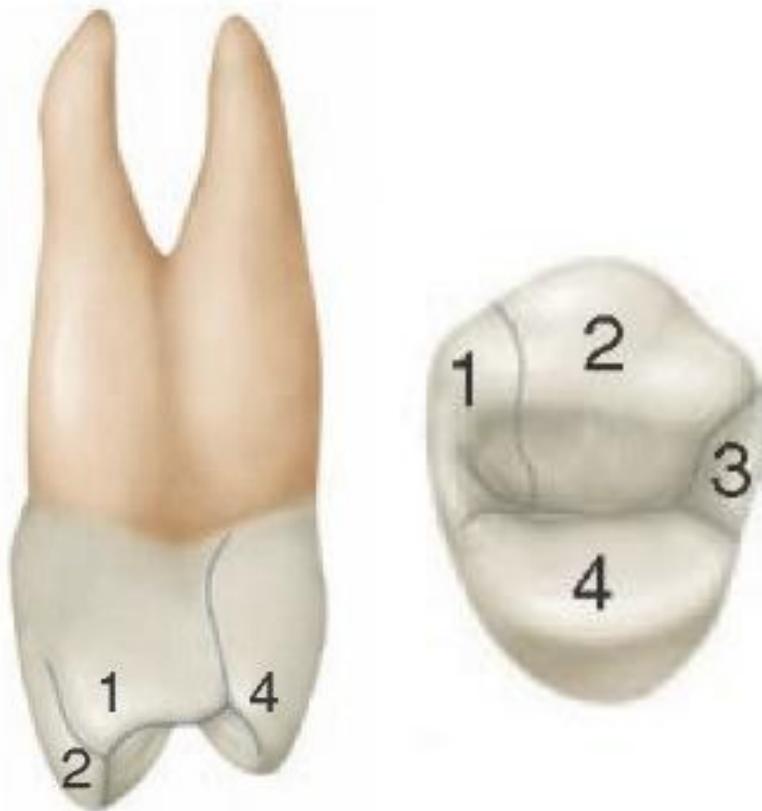


➤ Each tooth develops from 4 lobes or more.

➤ Anteriors: 4 lobes
(3 Labial & 1 Lingual)

Lobe

One of the primary centers of calcification and growth formed during crown development.

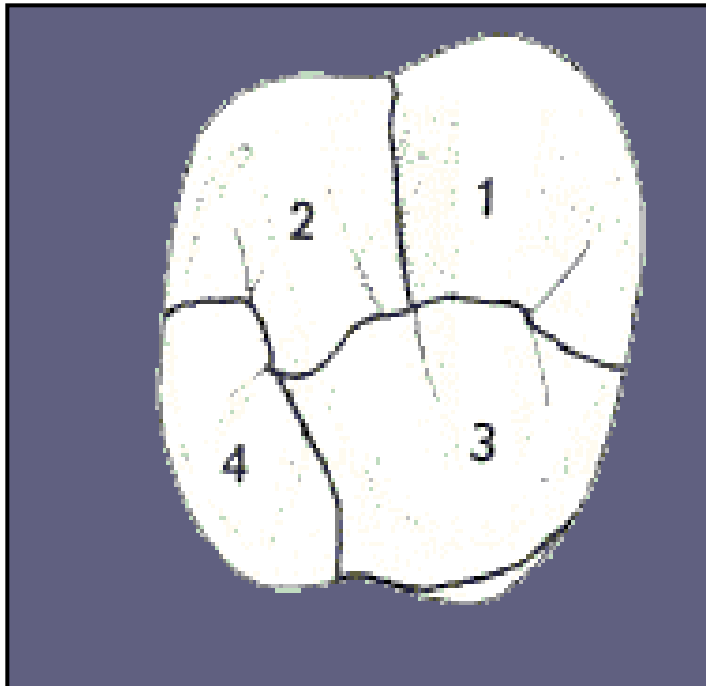


➤ Each tooth develops from 4 lobes or more.

➤ Premolars: 4 lobes
(3 Buccal & 1 Lingual)

Lobe

One of the primary centers of calcification and growth formed during crown development.



- **Each tooth develops from 4 lobes or more.**
- **Molars: 4 lobes
(2 Buccal & 2 Lingual)**

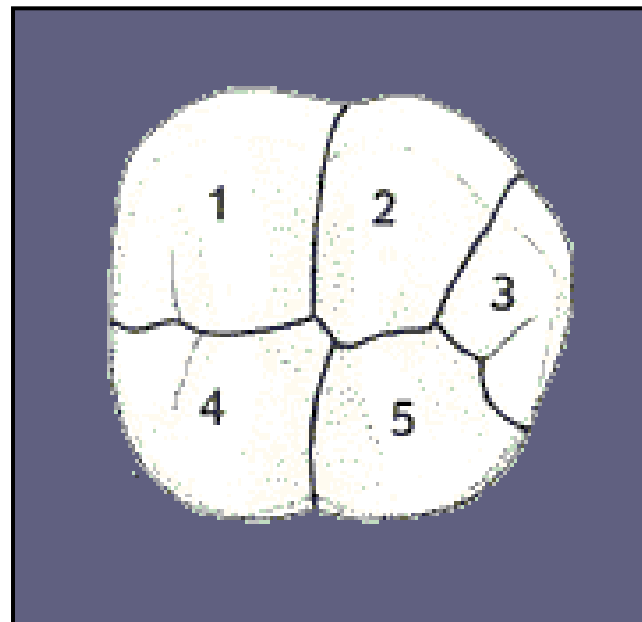
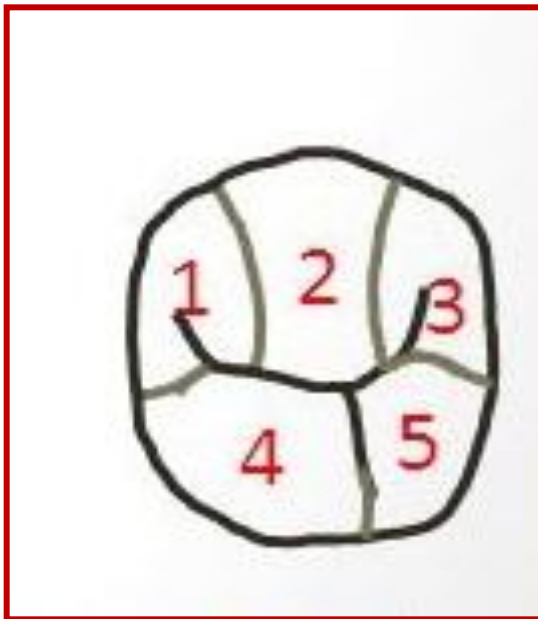
Five lobes may be found in:

- **Lower second premolar (3 cusp type)**

(3 Buccal & 2 Lingual)

- **Lower first molar**

(3 Buccal & 2 Lingual)

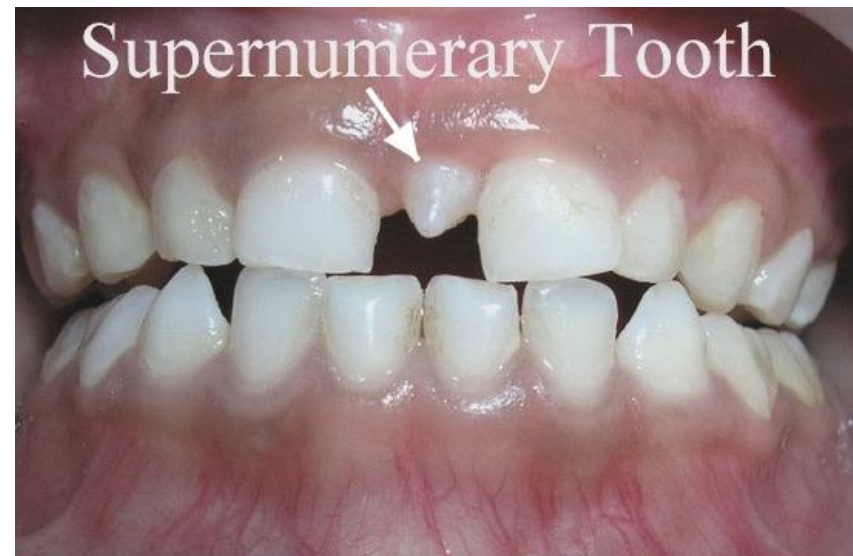


Less than 3 lobes may be found in:

- **Peg-shaped upper lateral incisor.**



- **Supernumerary teeth.**



1.Lobe

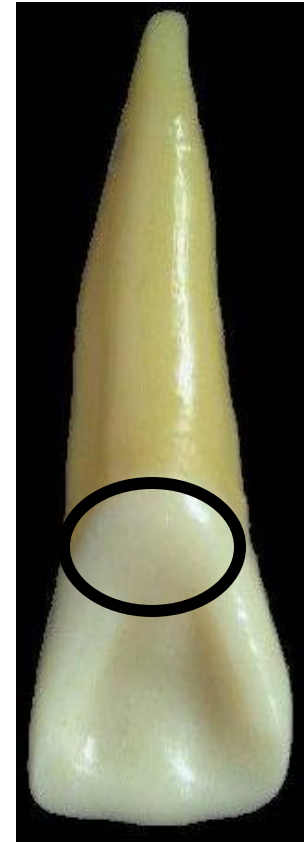
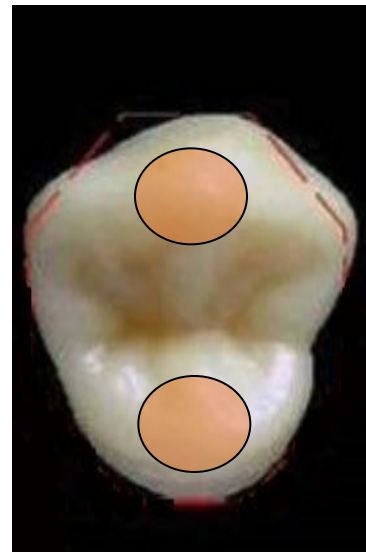
Mature form

Mamelons



Cingulum

Cusps

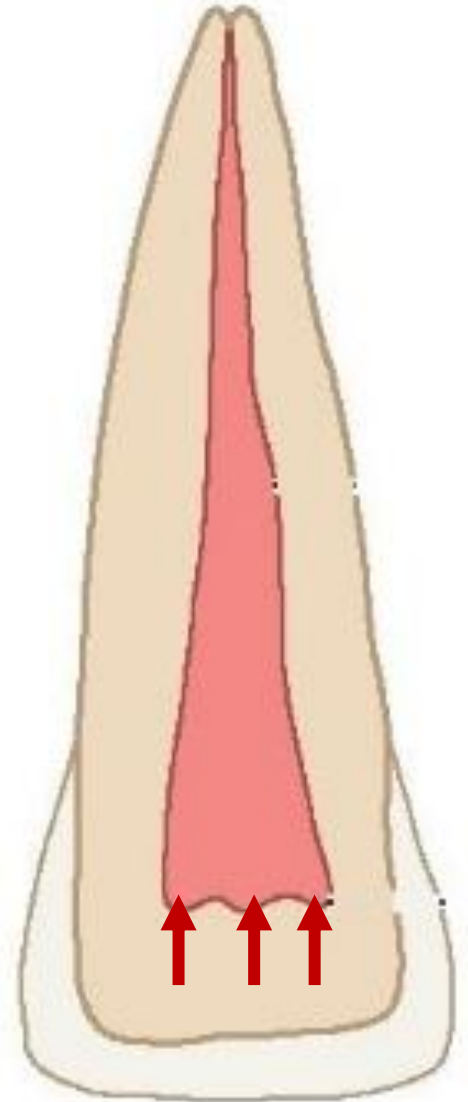


A. Mamelons

- **Three** small round projections of enamel present in the incisal third of **newly erupted incisors**.
- After normal use, they eventually **wear down** into a flat edge.

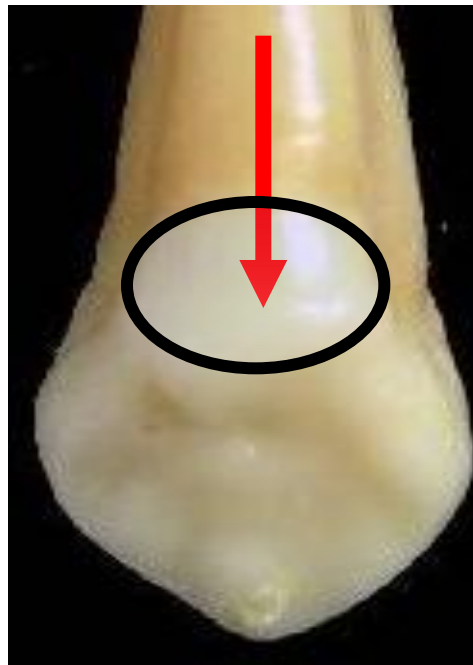


- Pulp chamber of upper incisors has **3 pulp horns** corresponding to these **3 labial lobes** (mamelons).



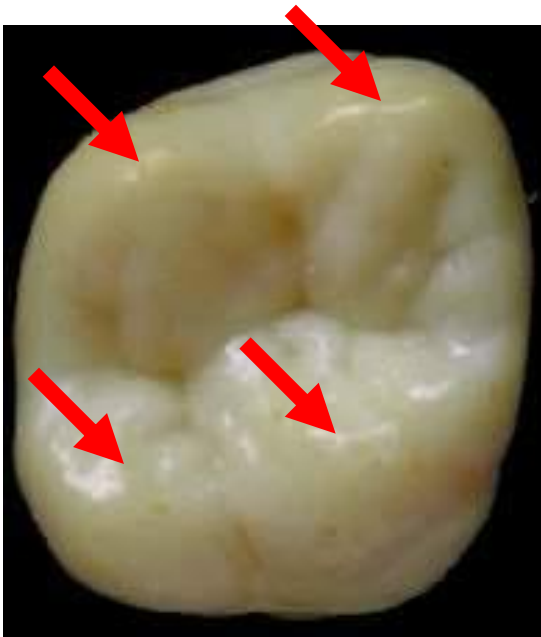
B. Cingulum

It is a **rounded prominence or bulge** on the **cervical third** of the **lingual surface** of the crown of anterior teeth.



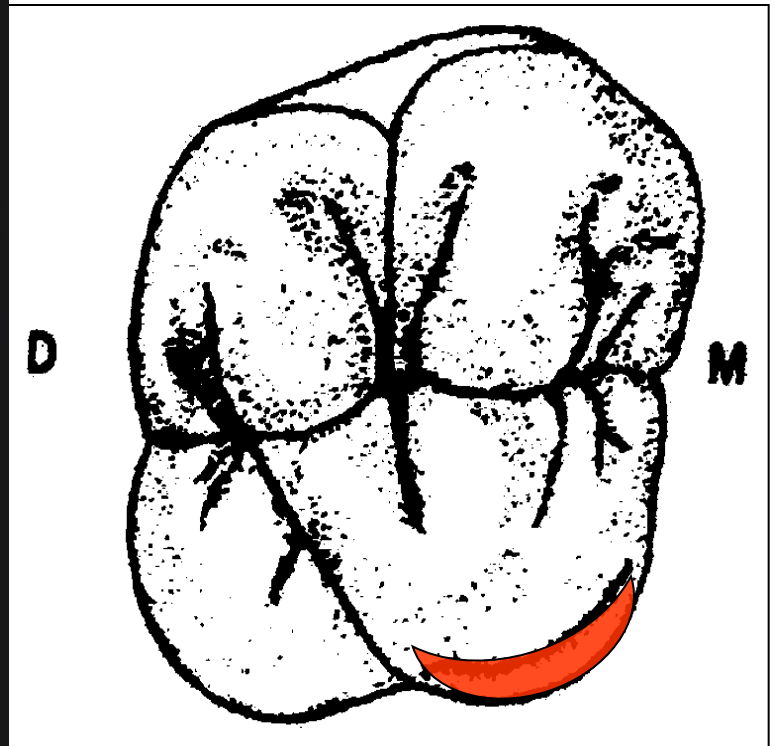
C. Cusp

- An elevation nearly **pyramidal** in shape.
- Formed of enamel, dentin & pulp.
- Found on **occlusal surface** of posterior teeth & incisal portion of canine.



2.Tubercle

- It is a **small elevation**, produced by excessive formation of enamel **ONLY**.
- CUSP  TUBERCLE



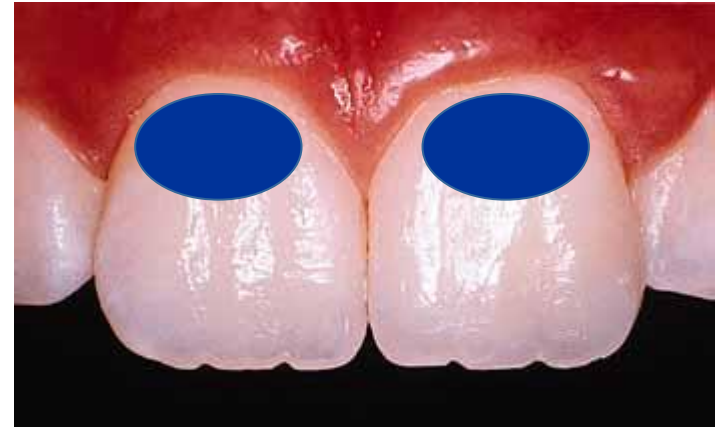
Ridge

Definition: It is a *linear elevation* on the different surfaces of the crown.

1- **Ridges** can be found on the Facial surfaces:

A- Labial (that exist in canines) or Buccal (that exist in premolars).

B- Cervical: in all teeth at the cervical
1/3



2- **Ridges** can be found on the Lingual surfaces:

A- Lingual (that exist in canines).

B- Marginal: in all anterior teeth.

C- Incisal: in all anterior teeth.



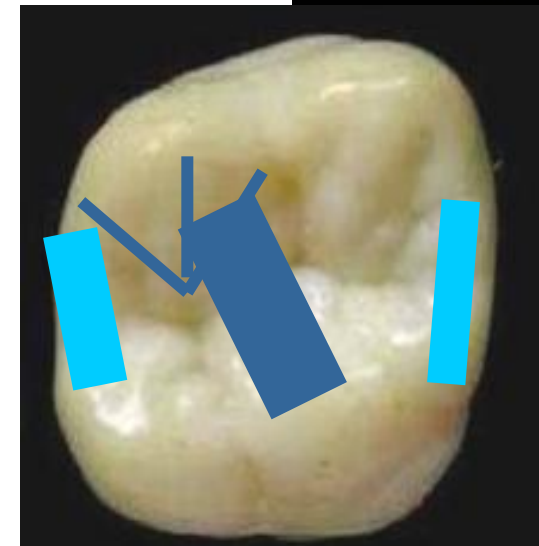
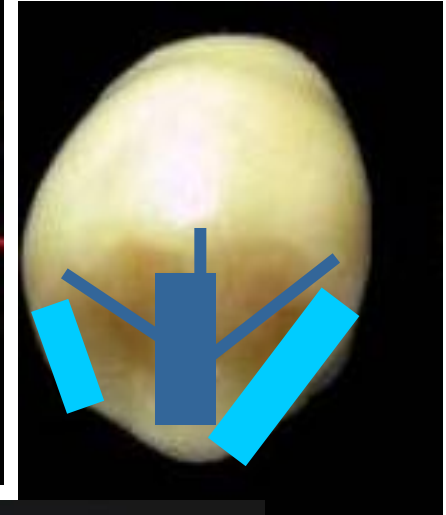
3- **Ridges** can be found on the occlusal surfaces:

A- **Triangular** (that exist in all posterior teeth).

B- **Marginal**: in all posterior teeth.

C- **Transverse**: in lower premolars.

D- **Oblique**: in upper molars.



Triangular ridge

In all posterior teeth.

Cusp ridge

In all posterior teeth.



Transverse ridge

In lower premolars.

Oblique ridge

In upper molars.



